Case-III :~ Equations solvable for x :

If the given equation, takes the form

$$\rightarrow \boxed{x = f(y, p)} \quad \text{——— (1)} \qquad \{\text{function of } y \text{ and } p\}$$

then differentiate w.r.t y gives an equation of the form

$$\rightarrow \frac{dx}{dy} = \frac{1}{p} = \phi\left(y, p, \frac{dp}{dy}\right) \qquad \left\{\text{function of } y, p, dp/dx\right\}$$

Now, solve this new differential equation in y and p

$$\left\{\text{Note} \rightarrow \frac{dx}{dy} = \frac{1}{p}\right\} \Rightarrow \frac{dx}{dy} = \frac{1}{dy/dx} = \frac{1}{p}$$

→ here, eliminate the value of p and put this value of p in eq-(1)

→ In case, eliminating of p is not possible then eliminate the value of y and put this value of y in eq-(1) to get the general solution.

Akash

PYQ-2021

Q.1. Solve;

$$y = 2px + y^2p^3$$

⇒ Given;

$$y = 2px + y^2p^3$$

$$\Rightarrow 2px = y - y^2p^3$$

$$\Rightarrow x = \frac{y - y^2p^3}{2p} \quad \text{—— (1)}$$

It is in the form of $x = f(y, p)$

∴ differentiate w.r.t y both side

$$\Rightarrow \frac{dx}{dy} = \frac{1}{p} = \frac{p(1 - 2yp^3 - y^2 3p^2 dp/dy) - (y - y^2p^3) dp/dy}{2p^2}$$

$$\Rightarrow \frac{1}{p} = \frac{p(1 - 2yp^3 - y^2 3p^2 dp/dy) - y\, dp/dy + y^2p^3 dp/dy}{2p^2}$$

$$\Rightarrow 2p = p - 2p^4y - y^2 3p^3 dp/dy - y\, dp/dy + y^2p^3 dp/dy$$

$$\Rightarrow -p - 2p^4y - y^2 3p^3 dp/dy - y\, dp/dy + y^2p^3 dp/dy = 0$$

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$\Rightarrow -p - 2p^4y - 2y^2p^3\, dp/dy - y\, dp/dy = 0$

$\Rightarrow -2p^3y\,[p + y\, dp/dy] - 1\,[p + y\, dp/dy] = 0$

$\Rightarrow [p + y\, dp/dy]\,[\underbrace{-1 - 2p^3y}_{\text{Singular solution}}] = 0$

$\therefore\ p + y\, dp/dy = 0$

$\Rightarrow p = -y\, dp/dy$

$\Rightarrow \frac{1}{y}\, dy = -\frac{1}{p}\, dp$

Integrate both side

$\Rightarrow \int \frac{1}{y}\, dy = \int -\frac{1}{p}\, dp$

$\Rightarrow \log y = -\log p + \log c$

$\Rightarrow \log y + \log p = \log c$

$\Rightarrow \log(py) = \log c$

$\Rightarrow py = c$

$\Rightarrow p = c/y$

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putting the value of $p = c/y$ in eq (1), we have

$$x = \frac{y - y^2 \times c^3/y^3}{2c/y}$$

$$= \frac{y\left[y - c^3/y\right]}{2c}$$

$$x = \frac{y^2 - c^3}{2c} \quad //$$

Q.2. Solve;

$$p^3 - 4xyp + 8y^2 = 0$$

=> Given;

$$p^3 - 4xyp + 8y^2 = 0$$

$$4xyp = p^3 + 8y^2$$

$$x = \frac{p^3 + 8y^2}{4yp} \quad \text{——} \quad (1)$$

$$x = \frac{p^2}{4y} + \frac{2y}{p}$$

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It is in the form of $x = f(y, p)$

differentiate both side w.r.t y

$$\frac{dx}{dy} = \frac{1}{p} = \frac{(3p^2 dp/dy + 16y)yp - (p^3 + 8y^2)[p + y\,dp/dy]}{4y^2p^2}$$

$$\Rightarrow \frac{1}{p} = \frac{3yp^3 dp/dy + 16y^2p - p^4 - 8y^2p - p^3y\,dp/dy - 8y^3\frac{dp}{dy}}{4y^2p^2}$$

$$\Rightarrow 4y^2p = 3yp^3 dp/dy + 8y^2p - p^4 - p^3y\,dp/dy - 8y^3 dp/dy$$

$$\Rightarrow 3yp^3 dp/dy + 4y^2p - p^4 - p^3y\,dp/dy - 8y^3 dp/dy = 0$$

$$\Rightarrow 4y^2p - p^4 + 2p^3y\,dp/dy - 8y^3 dp/dy = 0$$

$$\Rightarrow p^3[2y\,dp/dy - p] - 4y^2[2y\,dp/dy - p] = 0$$

$$\Rightarrow [2y\,dp/dy - p]\underbrace{[p^3 - 4y^2]}_{\text{singular soln}} = 0$$

$$\therefore 2y\,dp/dy - p = 0$$

$$\Rightarrow 2y\,dp/dy = p$$

$$\Rightarrow 2y\,dp/dy = p$$

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$$\Rightarrow \frac{2}{p}\,dp = \frac{1}{y}\,dy$$

Integrate both side

$$\Rightarrow \int \frac{2}{p}\,dp = \int \frac{1}{y}\,dy$$

$$\Rightarrow 2\log p = \log y + \log c$$

$$\Rightarrow \log p^2 = \log yc$$

$$\Rightarrow p^2 = yc$$

$$\Rightarrow p = \sqrt{yc}$$

putting the value of p in eq-(1)

$$x = \frac{yc\sqrt{yc} + 8y^2}{4yx\sqrt{yc}}$$

$$x = \frac{(yc)^{3/2} + 8y^2}{4y^{3/2}\sqrt{c}} \quad //$$

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Q.3. solve;

$$p^3y + 2px = y.$$

=> Given;

$$p^3y + 2px = y$$

$$2px = y - p^3y$$

$$x = \frac{y - p^3y}{2p}$$

It is in the form of $x = f(y, p)$

differentiate both side w.r.t y, we have

$$\frac{dx}{dy} = \frac{1}{p} = \frac{p\left(1 - [p^3 + 3p^2y\, dp/dy]\right) - [y - p^3y] \cdot dp/dy}{2p^2}$$

$$\Rightarrow \frac{1}{p} = \frac{p - p^4 - 3p^3y\, dp/dy - y\, dp/dy + p^3y\, dp/dy}{2p^2}$$

$$\Rightarrow 2p = p - p^4 - 3p^3y\, dp/dy - y\, dp/dy + p^3y\, dp/dy$$

$$\Rightarrow -p - p^4 - 2p^3y\, dp/dy - y\, dp/dy = 0$$

$$\Rightarrow -p^3[p + 2y\, dp/dy] - 1[p + y\, dp/dy] = 0$$

I think some error in question, if not please check?

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Q.4. solve;

$$p = \tan\left[x - \frac{p}{1+p^2}\right]$$

Sol Given;

$$\tan^{-1} p = x - \frac{p}{1+p^2}$$

$$\Rightarrow \quad x = \frac{p}{1+p^2} + \tan^{-1} p \quad \text{——— (1)}$$

It is in the form of $f(y, p)$

differentiate both side w.r.t y

$$\Rightarrow \frac{dx}{dy} = \frac{1}{p} = \frac{1}{1+p^2} \cdot \frac{dp}{dy} + \frac{dp/dy(1+p^2) - (2p\,dp/dy)\cdot p}{(1+p^2)^2}$$

$$\Rightarrow \frac{1}{p} = \frac{1}{1+p^2}\frac{dp}{dy} + \frac{dp/dy + p^2\,dp/dy - 2p^2\,dp/dy}{(1+p^2)^2}$$

$$\Rightarrow \frac{1}{p} = \frac{(1+p^2)\,dp/dy + dp/dy + p^2\,dp/dy - 2p^2\,dp/dy}{(1+p^2)^2}$$

$$\Rightarrow \left(\frac{1+p^2}{p}\right)^2 = dp/dp + \cancel{p^2\,dp/dy} + dp/dy + \cancel{p^2\,dp/dy} - 2\cancel{p^2\,dp/dy}$$

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$$\Rightarrow \left(\frac{1+p^2}{p}\right)^2 = \frac{2\,dp}{dy}$$

$$\Rightarrow \quad dy = \frac{2p}{(1+p^2)^2}\,dp$$

Integrate both side

$$\Rightarrow \int dy = \int \frac{2p}{(1+p^2)^2}\,dp$$

here $\int \frac{2p}{(1+p^2)^2}\,dp$

let $z = 1+p^2$

$dz = 2p\,dp$

$dp = \frac{1}{2p}\,dz$

$$\int \frac{2p}{z^2} \cdot \frac{1}{2p}\,dz = \int \frac{1}{z^2}\,dz$$

$$= -\frac{1}{z}$$

$$= \frac{-1}{1+p^2}$$

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$\therefore \Rightarrow \quad y = \dfrac{-1}{1+p^2} + c$

$$\frac{1}{1+p^2} = c - y$$

$$1+p^2 = \frac{1}{c-y}$$

$$p = \sqrt{\frac{1}{c-y} - 1}$$

~~putting this value of $\frac{1}{1+p^2}$ in eq -①~~

putting this value of p in eq -①

$$x = \sqrt{\frac{1}{c-y} - 1} \times (c-y) + \tan^{-1}\sqrt{\frac{1}{c-y} - 1}$$

//.

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